

Beyond Frame: A Remediation Authoring Tool for Transforming 2D Artworks into VR Experiences

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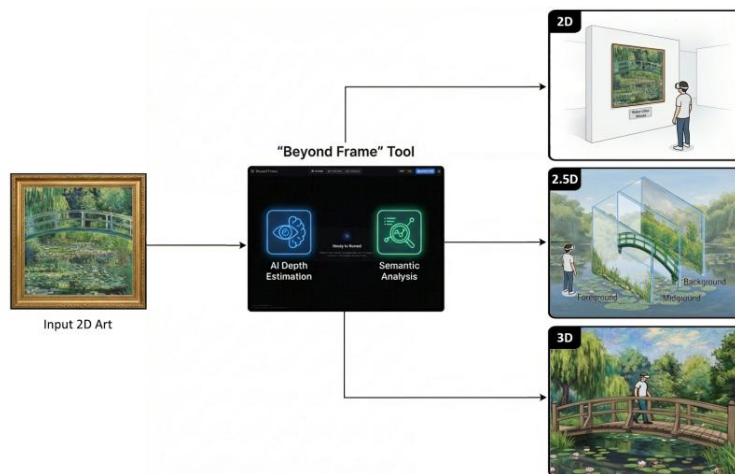


Figure 1: Workflow of Beyond Frame, transforming 2D art into 2D, 2.5D, and 3D VR experiences.

ABSTRACT

The rapid advancement of generative AI has expanded the possibilities for VR content creation, yet the remediation of 2D paintings into immersive 3D experiences remains a fragmented and technically demanding process. Creators often struggle with disjointed AI tools that produce randomized, uncontrollable results. This paper introduces a unified "2D-to-VR" authoring framework designed to lower technical barriers while creating a "Human-in-the-loop" workflow. We present a prototype system that automates the analysis and spatialization of 2D art into 2D, 2.5D, and 3D VR modes, and we demonstrate its efficacy through a pilot implementation using Claude Monet's *Water Lilies*.

Index terms: VR Remediation, Generative AI, Experiential Media, Immersive Content.

1 INTRODUCTION

In the era of spatial computing, transforming 2D legacy content into immersive VR experiences (Remediation) is a significant challenge. While generative AI tools like text-to-3D or skybox generators exist, they operate in silos. A creator must generate assets in one

tool, fix textures in another, and integrate them in a game engine—a workflow requiring high technical expertise. Furthermore, existing "one-click" solutions often suffer from hallucinations, producing 3D scenes that deviate significantly from the original artwork's narrative. Drawing upon recent research in interactive storytelling [1, 3, 4], VR user experience [2, 5], and digital heritage [6], our research addresses this gap by proposing an integrated authoring tool. We aim to provide a streamlined pipeline that utilizes AI for labor-intensive tasks (segmentation, modeling) while ensuring the creator retains control through a dedicated editing interface. This demo presents our initial prototype, capable of converting a single 2D painting into a multi-modal VR experience.

2 PROPOSED FRAMEWORK: 2D-TO-VR PIPELINE

The proposed framework operates on a four-stage semi-automated pipeline:

2.1 Image Analysis (Input)

The system analyzes the input painting's metadata (Artist, Era) for context. Simultaneously, it employs computer vision models (e.g., Monocular Depth Estimation) to decompose the image into semantic layers (foreground, midground, background) and predict the 3D spatial structure.

2.2 Asset Generation & Scene Construction

Based on the analysis, the system procedurally generates the scene.

- **2D Mode:** Automatic framing in a virtual gallery.
- **2.5D Mode:** Layer-based diorama construction using textured billboards.
- **3D Mode:** Volumetric reconstruction where recognized elements (e.g., "Bridge") are replaced with generated 3D models and standard assets, matched to the original composition.

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2.3 User Editing (UX)

To mitigate the stochastic nature of AI generation, we incorporate a "Human-in-the-loop" editing interface. This system comprises a Layer Editor for depth adjustment, precise Scale Modification tools, and Prompt Refinement for texture regeneration.

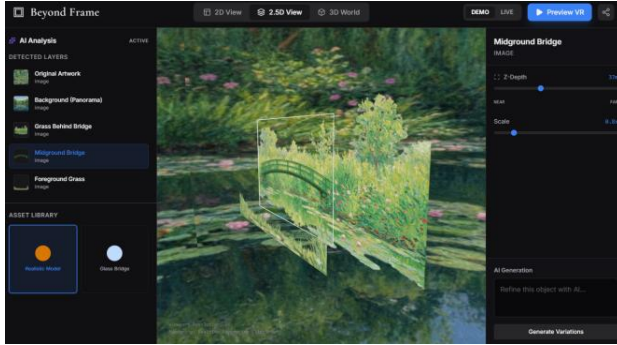


Figure 2: User Editing interface in 2.5D mode.

2.4 Multi-Mode Output

The final scene is exported as a scalable WebXR experience, offering three levels of immersion: 2D Gallery, 2.5D Portal, and 3D Free-Room.

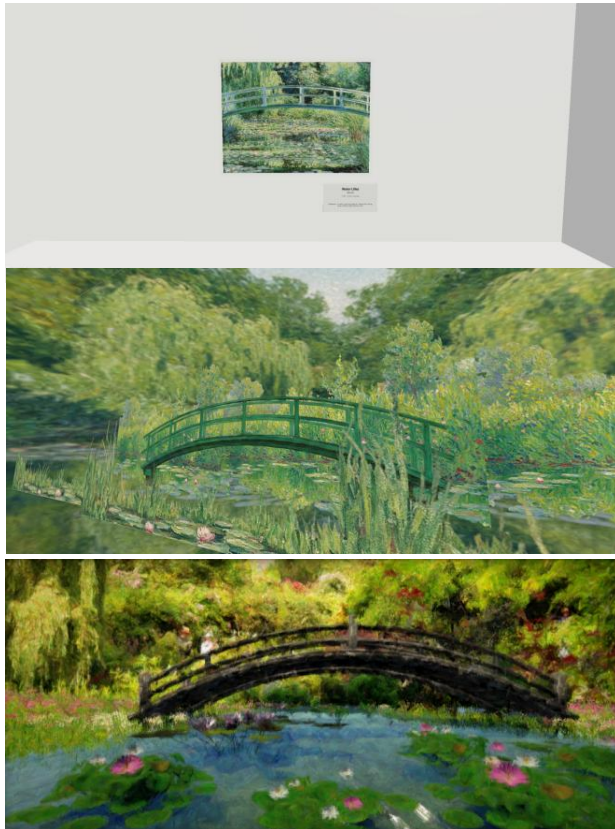


Figure 3: VR view of 2D, 2.5D, and 3D content.

3 CASE STUDY

To validate the feasibility of our framework, we implemented a functional prototype and conducted a case study using Claude Monet's *Water Lilies*.

- **Implementation:** The prototype was built using WebXR and client-side AI models (Xenova/transformers). This successfully demonstrated that a single painting could be decomposed into depth layers and reconstructed as a walkable 3D environment in real-time.
- **Observations:** During the automated segmentation process, issues such as empty spaces in 2.5D layers and skybox seams were discovered. Additionally, problems arose with the generation of 3D assets that did not match the painting style. These technical pain points became the decisive factor for including specific correction features like the 'Asset Library' and 'Style Filter' in the authoring tool UI.

4 CONCLUSION

This demo highlights the potential of an integrated AI-assisted workflow for VR remediation. Our pilot implementation confirms that technical barriers can be significantly lowered by automating the "2D-to-Spatial" conversion process. Future work will follow a Research through Design methodology, iterating on the prototype based on user feedback to refine the UX, and conducting formal user studies with media artists to evaluate the tool's usability and the expansion of aesthetic experience facilitated by the proposed method.

ACKNOWLEDGMENTS

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VIDEO URL

<https://youtu.be/VByFzBZ7Wko>

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